



Machine learning prediction and optimization of bio-oil production from hydrothermal liquefaction of algae

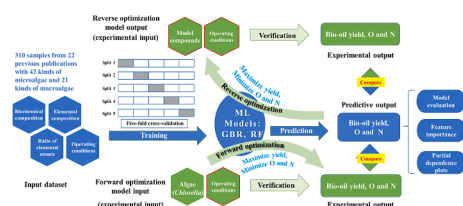
Weijin Zhang^{a,1}, Jie Li^{c,1}, Tonggui Liu^b, Songqi Leng^b, Lihong Yang^a, Haoyi Peng^a, Shaojian Jiang^a, Wenguang Zhou^b, Lijian Leng^{a,*}, Hailong Li^a

^a School of Energy Science and Engineering, Central South University, Changsha 410083, PR China

^b Key Laboratory of Poyang Lake Environment and Resource Utilization, Ministry of Education, and School of Resources, Environmental & Chemical Engineering, Nanchang University, Nanchang, Jiangxi 330031, PR China

^c Department of Chemical and Biomolecular Engineering, National University of Singapore, Singapore 117585, Singapore

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:

Algal biomass
Hydrothermal liquefaction
Biocrude oil
Machine learning
Prediction and optimization

ABSTRACT

Hydrothermal liquefaction (HTL) of algae is a promising biofuel production technology. However, it is always difficult and time-consuming to identify the best optimal conditions of HTL for different algae by the conventional experimental study. Therefore, machine learning (ML) algorithms were applied to predict and optimize bio-oil production with algae compositions and HTL conditions as inputs, and bio-oil yield (Yield_{oil}), and the contents of oxygen (O_{oil}) and nitrogen (N_{oil}) in bio-oil as outputs. Results indicated that gradient boosting regression (GBR, average test $R^2 \sim 0.90$) exhibited better performance than random forest (RF) for both single and multi-target tasks prediction. Furthermore, the model-based interpretation suggested that the relative importance of operating conditions (temperature and residence time) was higher than algae characteristics for the three targets. Moreover, ML-based reverse and forward optimizations were implemented with experimental verifications. The verifications were acceptable, showing great potential of ML-aided HTL for producing desirable bio-oil.

1. Introduction

Algae, grow quickly by consuming carbon dioxide in the atmosphere and nutrients (treat pollutants) in water (wastewater), are promising

candidates to mitigate climate change, energy depletion, and environmental pollution (Leng et al., 2020a; Leng et al., 2018b; Savage, 2012). The cultivation and processing of algae for such purposes have been carried out for several decades. Many promising technologies have been

* Corresponding author.

E-mail address: ljchs@126.com (L. Leng).

¹ These authors contributed equally to this work.

developed for converting algae biomass into alternative biofuels such as bio-oil. For example, hydrothermal liquefaction (HTL), carried out under sub-/supercritical water, is energy efficient and preferable to produce bio-oil from algae as it exempts the energy-intensive biomass drying process (Huang and Yuan, 2015; Leng et al., 2015).

However, the bio-oil produced from HTL of algae has inferior fuel qualities such as high contents of oxygen (O_{oil}) and nitrogen (N_{oil}) in bio-oil, which would cause the low energy density and the emission of nitrogen (N) oxides after combustion (Leng et al., 2018a). According to our previous review, 20% to 40% of the N in the raw biomass distributes into the oil phase, contributing to N_{oil} being up to 10 wt% with average values of 5–7 wt% and 3–6 wt% for microalgae and macroalgae, respectively (Leng et al., 2020c). The high N_{oil} from microalgae is mainly from protein (or N) in microalgal biomass (N content 6–9 wt%) and carbohydrate in the biomass (L. Leng et al., 2020c; Xu et al., 2021). In addition, the content of oxygenated chemicals (i.e., acids, alcohols, esters, and ketones) derived from biomass components result in high O_{oil} (Song et al., 2019). In general, the O_{oil} of macroalgae is higher than microalgae during HTL (Madsen et al., 2017). The two components (namely protein and carbohydrate) contribute significantly to the bio-oil yield ($Yield_{oil}$), O_{oil} , and N_{oil} from complicated reactions in HTL such as Maillard reactions (Peterson et al., 2008; Sohail et al., 2011). On the other hand, the HTL operating parameters such as temperature (T) and residence time (RT) also significantly affect $Yield_{oil}$, O_{oil} , and N_{oil} . Unfortunately, high $Yield_{oil}$ always accompanies high O_{oil} and N_{oil} content in most experimental studies (Fan et al., 2020; Leng et al., 2020c). Due to the complicated reactions and various variables, bio-oil production optimization, such as high $Yield_{oil}$, and low O_{oil} and N_{oil} , is challenging through simple traditional experimental methods.

Artificial intelligence, such as machine learning (ML), has recently been adopted in various biofuel production systems to promote the development of technologies such as hydrothermal conversions and pyrolysis to predict and optimize char, oil, and gas products (Li et al., 2021c, 2020a). Up to now, a few studies have been carried out to build models for predicting the yield and fuel characteristics of bio-oil from HTL or pyrolysis. Sheng et al. (2018) established a non-linear regression model to predict $Yield_{oil}$ from HTL of microalgae using model compounds. They achieved $R^2 > 0.98$ in the prediction model with 30 experimental data points. A comparison between the prediction performance of Random Forest (RF) and multiple linear regression was carried out to separately predict the yield and Hydrogen (H) content of bio-oil obtained from pyrolysis with biomass, and the RF performed better with R^2 0.80–0.90 (Tang et al., 2020). Furthermore, the yield of bio-oil from HTL of biomass was modeled using a decision support system, and the prediction accuracy was >94% (Gopirajan et al., 2021). Three ML models, namely Decision Regression Tree, RF, and Gradient Boosting Regression (GBR), were successfully applied to predict $Yield_{oil}$, N_{oil} , and energy recovery with R^2 0.69–0.80 (Li et al., 2021c). However, the prediction accuracy is unsatisfactory due to the significant divergence and complexity of biomass types and their compositions. In addition, most studies employed ML to conduct single-target prediction on bio-oil yield or properties and lack experimental verification.

To overcome the limitations mentioned above, we mainly focus on only one type of biomass (i.e., algae) to build ML models for increasing the predictive performance of the bio-oil property and downstream ML-based optimization for guiding experimental design. In detail, single and multi-target tasks GBR and RF models were established for predicting $Yield_{oil}$ and O_{oil} , and N_{oil} . Then, the predictive performance assessment and feature importance and partial dependence analysis were conducted to comprehend the HTL process. Moreover, to provide helpful guidance for aiding desired bio-oil production, two optimization strategies (reverse and forward optimizations) were implemented for providing ideal feedstock composition and corresponding HTL condition, as well as tailoring HTL conditions for specific alga (i.e., *Chlorella* in this work). Finally, the optimal solutions based on ML optimization were employed to guide HTL experiments for improving bio-oil properties (i.

e., improve $Yield_{oil}$ and reduce O_{oil} and N_{oil} in this work).

2. Material and methods

2.1. Dataset compilation and pre-processing

The dataset (310 samples) was compiled by collecting original experimental data from 22 previous publications. The keywords, including hydrothermal liquefaction, microalgae, macroalgae, algae, and bio-oil, were searched from the literature databases such as Web of Science and Scopus. The data containing algal feedstock compositions and operational parameters were gathered from the tables or extracted from the plots by PlotDigitizer in related literature and corresponding supplementary materials. There were 42 types of microalgae and 21 types of macroalgae in the dataset (E-supplementary data of this work can be found in online version of the paper). In detail, input parameters in each dataset include elemental composition (i.e., C, H, O, N and S), the atomic ratio of elements (H/C, O/C, and N/C), the biochemical composition of algae (such as protein, lipid, carbohydrate, and ash), and HTL operating conditions (temperature, residence time, and solid content of algae). The output parameters of the HTL process, such as $Yield_{oil}$ and elementary composition of bio-oil, were collected or calculated on a dry basis. To avoid inconsistency of units and improve the comparability of the dataset, all units of each feature in the dataset were uniformed.

Therein, the elementary and biochemical compositions of algae biomass in datasets were uniformed by using the following two equations, respectively:

$$O = 100 - C - H - N - S - \text{Ash} \quad (1)$$

$$\text{Protein} + \text{Lipid} + \text{Carbohydrate} + \text{Ash} = 100 \quad (2)$$

where O, C, H, N, and S represent the values of oxygen content (wt%), carbon content (wt%), hydrogen content (wt%), nitrogen content (wt%), and sulfur content (wt%) in algae, respectively.

The elemental compositions of bio-oil in datasets were uniformed as followed:

$$O_{oil} = 100 - C_{oil} - H_{oil} - N_{oil} \quad (3)$$

where O_{oil} , C_{oil} , H_{oil} , and N_{oil} indicate the values of oxygen content (wt%), carbon content (wt%), hydrogen content (wt%), and nitrogen content (wt%) in bio-oil, respectively.

For achieving a uniform range among the variables, all the input values were normalized using Eqs. (3):

$$x_i^* = \frac{x_i - \mu}{s} \quad (4)$$

where x_i^* represents the standardized value of each input variable, and x_i represents the original experimental value of the input variable. μ and s represent the mean value and standard deviation of x_i , respectively.

Before training ML models, the dataset was randomly divided into 80% data as the training dataset and the remaining 20% data as the testing dataset to evaluate the trained ML models (Li et al., 2007).

2.2. Machine learning model algorithms and hyper-parameter optimization

Two classical kinds of tree-based ML algorithms, Random Forest (RF) and Gradient Boosting Regression (GBR), were applied to train predictive models in this work. They expressed many advantages, such as handling high-dimension datasets with unnecessary dimension reduction and feature selection, strong interpretability, fast convergence during training, few hyper-parameters that need to be specified, and good compatibility for unbalance datasets (Li et al., 2021c). Moreover, some of the published works have reported that these two algorithms showed good potential in the biomass valorization domain due to their strong ability in modeling for classification and regression to thermal conversion process (Li et al., 2021a, 2020a; Zhu et al., 2019).

RF is an ensemble learning strategy that develops and combines many decision trees to achieve accurate prediction and improved generalization results compared to single decision tree models. During the algorithms learning process, hundreds of decision trees are built with the input dataset by bagging or bootstrap aggregation. Then, each tree is a regressor, and the final model prediction result for RF is the average output value of all individual decision trees. Two essential hyper-parameters, the number of trees ($n_{\text{estimators}}$) and the maximum depth of trees (max_depth), were optimized in the model training and developing process of this study.

GBR algorithm is trained using a boosting method, another ensemble learning algorithm (Natekin and Knoll, 2013). First, a leaf is achieved from the mean of the observed output values, which is also an initial prediction value for obtaining the initial errors (Li et al., 2020a). Then gradient boost strategy structures the first regressor tree to predict the errors according to the change between actual and original values. Next, the new predicted value is gained by adding the previous predicted value and the prediction error of the new regressor tree (Li et al., 2021b). Finally, the next regressor tree is further built from the new residual of predicted values of the prior tree and actual experimental values until the residuals are maximally reduced. In the present study, the tuned important hyper-parameters were $n_{\text{estimators}}$ and the max_depth .

The 5-fold cross-validation strategy was employed for optimizing two important hyper-parameters of the above ML models to improve their prediction ability during training (Fig. 1). In detail, the training data points will be randomly split into five parts; four will be used to train the model to find optimal hyper-parameters for the ML model. The left one-fifth of the training data points will be used for validating the performance. This process will iterate five times to let all the five parts of the data be validated. Finally, the average performance from five validations will be employed to identify the best hyper-parameters for the ML model. The number of $n_{\text{estimators}}$ and the max_depth were adjusted from 2 to 128 in multiples of 2. And other hyper-parameters of GBR and RF models were kept at default values. Finally, the ML models with tuned hyper-parameters were retrained by the training data and established to evaluate generalization capability by the testing data.

2.3. Model development and evaluation

Both single-target and multi-target models were developed for predicting bio-oil yield and properties in HTL of algae in this work. To obtain training parameters of the single-task models for node splitting, minimizing square errors is a straightforward way for each splitting part. This method is called the diminishing variance, which helps determine which node should be split into child nodes and the split value for this feature. The formula of the variance of any node, the loss function (L), was shown as Eq. (5) (Lin et al., 2019):

$$L = \sum_{i=1}^N (y_i - \bar{y})^2 \quad (5)$$

where N represents the number of actual values in the node, and y_i indicates the i^{th} actual value; $\bar{y}$ indicates the mean value of the node and the predicted value of the child node; a favorable division for a node forms two child nodes where all y_i close to $\bar{y}$.

A total loss function (L_M) of the multi-task predictive model is specified as follow (Liu et al., 2017):

$$L_M = \sum_{t=1}^M \alpha_t L_t(y_n^t, \hat{y}_n^t) \quad (6)$$

where M is the number of tasks; α_t represents the weight factor for each task t , which is used to underline the degree of relative importance for this task, and all the targets in this work are given equal weight; L_M indicates the loss function for each task t ; y_n^t is the actual value for each task t and $\hat{y}_n^t$ is the predicted value for each task t . In the multi-task predictions, the bio-oil yield, O and N content of bio-oil from HTL are crucial targets we want in the prediction. The yield told us how much bio-oil could be produced, and the O and N contents showed the quality of this bio-oil during HTL from algae. It was hard to identify different weight factors for them. Therefore, we treated them equally in multi-task predictions and optimizations.

The determination coefficient (R^2), a statistical index, is employed to evaluate the model. Generally, R^2 represents the fitting degree of the relationship between input and output variables.

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i^* - \hat{y}_i)^2}{\sum_{i=1}^N (y_i^* - \bar{y})^2} \quad (7)$$

where N indicates the number of test samples; y_i^* and $\hat{y}_i$ indicate the i^{th} experimental value and corresponding predicted value, respectively; $\bar{y}$ indicates the mean value of the N experimental values.

In addition, the root mean squared error (RMSE) and mean absolute error (MAE) were calculated to evaluate prediction performance. The RMSE was an indicator of the degree of dispersion between the actual and predicted values. For MAE, the actual situation of predicted value error was measured without considering their weights. The RMSE and MAE are calculated by Eqs. (8) and (9), respectively:

$$RMSE = \sqrt{\frac{1}{N} \left(\sum_{i=1}^N (y_i^* - \hat{y}_i)^2 \right)} \quad (8)$$

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i^* - \hat{y}_i| \quad (9)$$

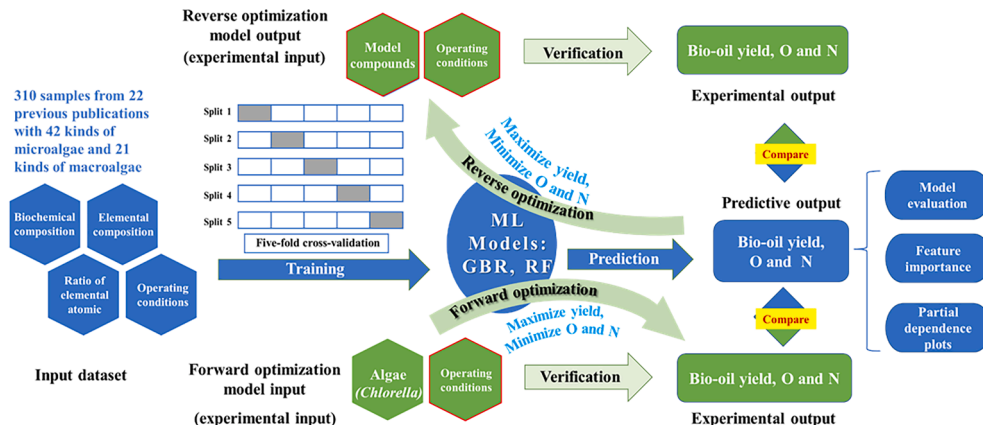


Fig. 1. Schematic diagram of ML-based prediction and optimization for desired bio-oil production in HTL of algae with experiment verification for comparison.

To distinguish the effects of different algae components on the accuracy of prediction models, the input variables were divided into four groups for training RF and GBR models: (i) all inputs except biochemical composition and the atomic ratio of elements (dataset #1); (ii) all inputs except elemental and biochemical composition (dataset #2); (iii) all inputs except elemental composition and the atomic ratio of elements (dataset #3), and (iv) all inputs (dataset #4). Then, all the models were established applying the *scikit-learn* library (scikit-learn.org/stable) in the Python programming platform.

2.4. Model optimization and experimental verification

2.4.1. Reverse optimization and experimental verification with model compounds

The multi-task GBR model was retrained based on the tuned hyper-parameters from dataset #3. Another valuable function of ML models is reverse optimization, which includes predicting an optimum point or a customized target point. For cases when only the outputs are concerned, the biomass biochemical composition and HTL parameters can be reversely optimized simultaneously. The reverse optimization was carried out by maximizing bio-oil yield and minimizing O and N contents (target) while feeding back corresponding biochemical biomass composition and HTL parameters for achieving the target. In the present work, particle swarm optimization (PSO) algorithm (Juneja and Nagar, 2016) was applied to conduct inverse optimization with searching from lower to an upper boundary (minimum to maximum values) of each input variable. At the beginning of the algorithm, the particles swarm (initial solution) is randomly initialized in the feasible solution space, where the position of each particle represents a solution of the problem, and the new solution is searched according to the objective function calculation. Then, at each iteration, the particle updates itself by tracking two values (the best solution searched by each particle and the best solution currently searched by the entire particle swarm). The maximum output of the PSO search corresponds to the input variables values, which are the local or global optimal solutions (Qi and Chen, 2018). The boundaries (minimum to maximum values) of the input variables were confirmed with the analysis results of box plots (see [supplementary material](#)). During optimization, the weighted sum method (one of the most widely used multi-objective optimization methods) was used to maximize the Yield_{oil} and minimize the O_{oil} and N_{oil} with an equal weighting factor. The reverse optimizations of feedstock characteristics and operating conditions can guide experiments through trained ML models and the PSO optimal algorithm. The soya protein, soybean oil, and wheat straw-derived cellulose have been widely used as model components of biomass, including algae (Qiao et al., 2019; Yang et al., 2018). Thus, they are used as model compounds, representing protein, lipid, and carbohydrate, respectively. The characteristics of model compounds composition are shown in Table 1. And the biomass characteristics and HTL parameters can be further used to produce bio-oil for experimental validation by comparing the Yield_{oil}, O_{oil}, and N_{oil} from experiments and models.

2.4.2. Forward optimization of bio-oil production from specified algae with experimental verification

The multi-task GBR model with tuned hyper-parameters was trained from dataset #4. After ML models were successfully developed, we aimed to validate whether the optimal solutions from modeling were feasible or not. Hence, we chose an easy-available alga, i.e., *Chlorella*, as a case study to investigate the forward optimization. Moreover, the biochemical compositions of the *Chlorella* are within the ranges of our collected dataset (see [supplementary material](#)). Hence, our developed model can recognize it well during the optimization process. For cases when one specific alga (i.e., *Chlorella* in this work) is given, only the HTL operating parameters need to be optimized for getting maximized Yield_{oil} and minimized N_{oil} and O_{oil} by inputting all possible variables within the model in specified step size, for example, step size of

Table 1

Characteristics of *Chlorella* and model compounds.

Items	<i>Chlorella</i>	Model compounds		
		soya protein	soybean oil	wheat straw-derived cellulose
Elementary composition ^b				
C (%)	47.68	46.29	76.70	41.80
H (%)	6.73	6.92	11.79	6.19
O (%) ^a	31.60	29.10	11.51	51.43
N (%)	10.88	13.52	0	0
S (%)	0.87	- ^c	—	—
The atomic ratio of elements				
H/C	1.694	1.794	1.845	1.777
O/C	0.497	0.472	0.113	0.923
N/C	0.171	0.250	0	0
Biochemical composition ^b				
Protein (%)	65.32	—	—	—
Lipid (%)	11.06	—	—	—
Carbohydrate (%)	14.35	—	—	—
Ash (%)	2.24	4.17	0	0.58

^a By difference, O = 100 - C - H - N - S - Ash

^b Dry basis

^c Not analyzed

5 °C for T from 150 °C to 450 °C, step size of 1 min for RT from 0 min to 60 min, and step size of 1% for solid contents (SC) from 3% to 30%. The iteration outputs (containing extensive results with different operating conditions and bio-oil properties) were obtained to provide data support for filtering the optimal solutions. The experimental validations were conducted from the optimal iteration outputs results and further guided the experiment to produce the desired bio-oil. In the HTL experiment validation, the feedstock (powdered *Chlorella*) was bought from Realin Biotechnology co., China. It was dried at 105 °C to constant weight and stored in an airtight glass container. The feedstock compositions of the microalgae are listed in Table 1. Besides, the practical operating details (i.e., the setup of the reactor, the set of the reaction procedure, the phase separation process, and the collection of products and analysis methods) followed the methods reported in our previous studies (Leng et al., 2018c; Leng et al., 2020d). In brief, the batch reactor was heated to target T at a rate of 4–5 °C/min, and the HTL reaction was maintained at constant target T for a given RT. When the reactor was cooled to room temperature, dichloromethane was added to the mixture. Then, the solids were collected through filtration, and the aqueous phase was centrifugally separated from the oil phase. At last, the bio-oil was extracted through rotary evaporation for dichloromethane removal.

3. Results and discussion

3.1. Statistical analysis of dataset before modeling

The statistical analysis of the collected dataset (see [supplementary material](#)) shows the minimum, average, median, and maximum values and quartile-basis box plots of input and output features. The C and O were two primary elements in algae, and the C ranged 22–65.62 wt%. Three more parameters, the molar ratio of elements atomic (i.e., H/C, O/C, and N/C), were used as input variables for the model development, as they significantly affect the quantity and quality of the bio-oil (Leng et al., 2020b, 2020c). Another fundamental input variable category was the biochemical composition of algae, which can be used to confirm algal biomass type. The protein (%) and carbohydrate (%) ranges were from 0% to upper boundaries of 86–99% in this study. These results indicated that the data collected were adequate to cover most algae species (see [supplementary material](#)). For the HTL condition, the T of HTL ranged 150–450 °C, the RT ranged 0–240 min, and the SC of algae ranged 3–30 wt%. For output targets, the Yield_{oil} (%) was 1–62%, O_{oil} (%) was 6.5–31.3%, and N_{oil} (%) was 0.2–10.35%.

To evaluate whether considered input variables can provide valuable

information for ML model development to predict the interested output bio-oil properties, we conducted the Pearson correlation analysis to understand the correlations between inputs and outputs roughly. The higher Pearson correlation coefficient means a higher correlation with output, and a minus value represents negatively related, vice versa. According to the Pearson correlation coefficients matrix and significance level of correlation coefficients (Fig. 2), the bio-oil yield had a stronger positive correlation with the T ($r = 0.54$, $p < 0.001$) and lipid ($r = 0.49$, $p < 0.001$) than other variables. Moreover, both O_{oil} and N_{oil} were significantly correlated ($p < 0.001$) with biomass O content, N content, O/C, N/C, T, and ash. However, O_{oil} was negatively correlated with temperature ($r = -0.63$), while a positive correlation was found between N_{oil} and temperature ($r = 0.53$). It can be seen that almost all inputs features (except S and H/C) expressed significant relationships with N_{oil} ($p < 0.001$). In general, the considered input variables are more or less related to the output bio-oil properties. Hence, these input features will provide useful information for the development of predictive models. Besides, the properties of bio-oil varied observably because of various algae species and HTL operating conditions. Thus, it was necessary to develop and optimize the ML models to identify and mediate the crucial factors for bio-oil production with desired properties.

3.2. Single-task ML prediction models

3.2.1. Hyper-parameter adjusting and optimization

Fig. 3 depicts the results of average R^2 based on 5-fold cross-validation for hyper-parameter adjusting of the RF and GBR models based on the randomly selected 80% data points from dataset #4. For both models, two hyper-parameters (i.e., $n_{estimators}$ and max_depth) were optimized. The average R^2 of the GBR model showed satisfactory results when $n_{estimators}$ was higher than 20 and max_depth was lower than 10 (Fig. 3a-c). The optimal max_depth of 4 and $n_{estimators}$ of 100, 64, and 128 for Figs. a, b, and c, respectively.

For RF (Fig. 3e-g), the average R^2 was improved with the increase of max_depth until it reached ~ 8 , while no evident elevation was observed with the incremental $n_{estimators}$. Moreover, the average R^2 showed no noticeable change when max_depth was > 8 . Therefore, moderate $n_{estimators}$ (32) and relatively small max_depth (8 or 16) were chosen to avoid the overfitting of predictive models.

According to the above hyper-parameters optimization processes, the hyper-parameters of ML models for other datasets (#1–3) with different input variables (details not shown). Still, the same output was optimized using 5-fold cross-validation (summarized in Table 2).

3.2.2. Model assessment and feature importance analysis

Table 2 showed the single-task prediction performance, including R^2 , RMSE, and MAE of each optimal ML model from different input variables (datasets #1–4). By comparing the testing accuracy of models for

different targets, the performance between RF and GBR models was close, both with R^2 of ~ 0.90 . The test R^2 for GBR models was slightly higher than that for RF, while the test RMSE and MAE for GBR were smaller than those for RF, respectively (Table 2). These findings were consistent with those observed in previous studies that GBR models can get better results than RF in most cases (Dietterich, 2000). Among the four datasets, the GBR Model with dataset #4 achieved a test R^2 of 0.90 and the smallest test RMSE and MAE (Table 2 and Fig. 2a-c). Models may exhibit better prediction ability when more related input parameters are considered (Pathy et al., 2020). The bio-oil prediction performance here was acceptable and reasonable compared with the results from previous studies (Cheng et al., 2020; Li et al., 2021d; Tang et al., 2020).

The predicted values from the GBR model with dataset #4 vs. actual values in the datasets (contain training and testing data) are plotted in Fig. 4a-c. Almost all the predicted points and actual training points concentrated around the $y = x$ straight line. And the closing of the points to the straight line indicates better prediction accuracy. Compared with training data points, the distribution density of test data points around $y = x$ was smaller. Due to the inherent properties of ML models, the ML models always express a better prediction performance of the training dataset than that of the testing dataset (Li et al., 2020b).

According to the GBR models, the feature importance analysis was implemented to assess which input variables were significant for predicting Yield_{oil}, O_{oil}, and N_{oil}. Fig. 4b-c show the ranking of feature relative importance with the GBR model built from dataset #4. It depicts that the feature importance of SC was much less than the other input operating condition variables (T and RT) for three targets prediction.

Additionally, the input features were divided into four categories (i.e., the elemental composition, the biochemical composition, the ratio of elemental composition, and operating conditions) to figure out the importance degree of each feature category. In detail, the T is the foremost variable and is followed by lipid content and RT for Yield_{oil} prediction (Fig. 4d). According to experimental results and theoretical mechanisms, it has been widely accepted that the lipid content is the main factor affecting Yield_{oil} (Huang and Yuan, 2015; Sohail et al., 2011). Meanwhile, operating conditions (especially T and RT) were still primary influence factors, accounting for 61% of importance to O_{oil} prediction (Fig. 4e). However, each of the other three categories of algae compositions accounted for $< 15\%$. In N_{oil} prediction, aside from T, protein played a vital role mainly because only the protein contains N (non-proteinaceous N is ignored) among the biochemical composition (Fig. 4f and Table 1). Furthermore, the other compounds can interact with proteins to promote/inhibit generating N_{oil}.

3.3. Multi-task ML prediction models

3.3.1. Model development with optimal hyper-parameters

The single-task predicting the Yield_{oil}, O_{oil}, and N_{oil} was insufficient to evaluate HTL conversion performance comprehensively.

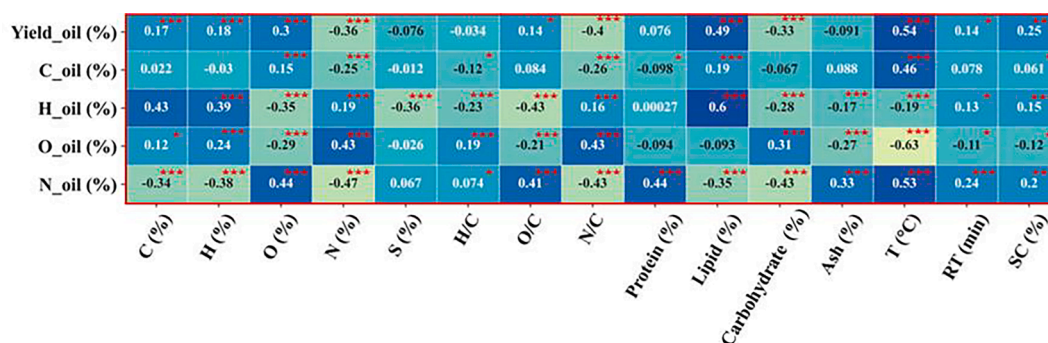


Fig. 2. Pearson correlation coefficients (r) matrix between inputs and outputs, expressing a positive or negative linear relationship (***: significance level $p < 0.001$, **: significance level $p < 0.01$, *: significance level $p < 0.05$).

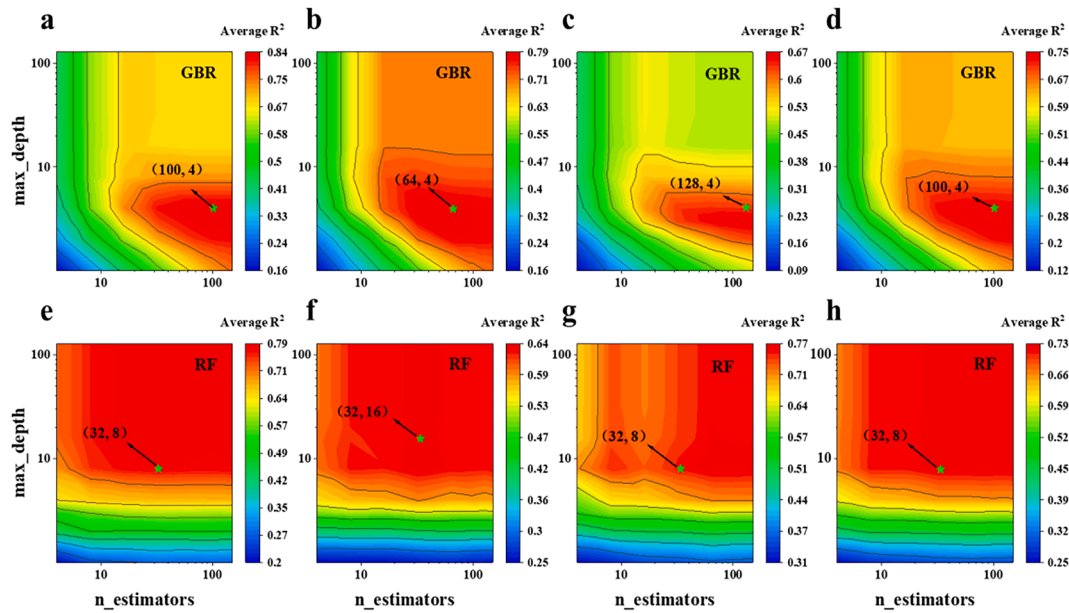


Fig. 3. Hyper-parameter tuning of ML models for predicting Yield_oil (a, e), O_oil (b, f), N_oil (c, g), and three targets together (multi-task) (d, h) based on dataset #4 (GBR: Gradient boosting regression, RF: Random forest).

Table 2

Summary of results from hyper-parameters adjusting and model performance of single-task ML prediction.

Dataset	Input variables	Target	Optimized model (n_estimators, max_depth)	Train R^2	Train RMSE	Train MAE	Test R^2	Test RMSE	Test MAE
#1	EC +Operating conditions	Yield_oil	GBR (128, 4)	0.96	2.73	1.78	0.90	4.75	3.59
			RF (32, 16)	0.95	2.99	2.03	0.86	6.06	4.52
		O_oil	GBR (100, 4)	0.91	1.63	0.92	0.89	1.71	1.23
			RF (32, 16)	0.90	1.67	0.95	0.87	1.85	1.35
		N_oil	GBR (128, 4)	0.97	0.41	0.27	0.88	0.77	0.56
			RF (32, 16)	0.94	0.53	0.34	0.87	0.78	0.57
#2	Ratios +Operating conditions	Yield_oil	GBR (128, 4)	0.96	2.82	1.92	0.88	5.07	3.72
			RF (32, 16)	0.95	3.24	2.19	0.85	5.08	3.92
		O_oil	GBR (100, 4)	0.90	1.68	1.01	0.87	1.88	1.40
			RF (32, 16)	0.89	1.77	1.03	0.86	1.97	1.49
		N_oil	GBR (100, 4)	0.89	0.42	0.29	0.87	0.80	0.55
			RF (16, 8)	0.90	0.70	0.46	0.86	0.82	0.60
#3	BC +Operating conditions	Yield_oil	GBR (128, 4)	0.96	2.73	1.78	0.90	4.75	3.59
			RF (32, 16)	0.95	2.99	2.03	0.86	6.06	4.52
		O_oil	GBR (100, 4)	0.91	1.63	0.92	0.89	1.71	1.23
			RF (32, 16)	0.90	1.67	0.95	0.87	1.85	1.35
		N_oil	GBR (128, 4)	0.97	0.41	0.27	0.88	0.77	0.56
			RF (32, 16)	0.94	0.53	0.34	0.87	0.78	0.57
#4	EC + BC +Ratios +Operating conditions	Yield_oil	GBR (100, 4)	0.96	2.69	1.74	0.90	4.69	3.56
			RF (32, 8)	0.93	3.17	2.39	0.87	5.72	4.34
		O_oil	GBR (128, 4)	0.92	1.55	0.82	0.90	1.68	1.23
			RF (32, 16)	0.90	1.68	0.97	0.87	1.91	1.43
		N_oil	GBR (64, 4)	0.96	0.47	0.31	0.90	0.68	0.53
			RF (32, 8)	0.93	0.61	0.42	0.88	0.74	0.56

EC: Elementary Composition (C, H, O, N and S); Ratios: Ratio of Elementary Composition (H/C, O/C, and N/C); BC: Biochemical Composition (Protein, Lipid; Carbohydrate and Ash); Yield_oil: The yield of bio-oil; O_oil: Oxygen Content in bio-oil; N_oil: Nitrogen Content in bio-oil; + denotes the combination of input variables under consideration.

Herein, the multi-target predicting these three targets was carried out to achieve the simultaneous prediction. Again, GBR and RF models were developed considering different compositions of algae and operating conditions as inputs, and Yield_oil, N_oil, and O_oil as the output. The hyper-parameter adjusting process of the GBR and RF for multi-task prediction is similar to single-task. The optimized n_estimators and max_depth of GBR based on dataset #4 were 100 and 4, and 32 and 8 for RF, respectively (Fig. 2d and h). After hyper-parameter adjusting for the ML models using 5-fold cross-validation, the final optimal models with their own adjusted hyper-parameters were obtained, as shown in Table 3.

To better figure out the predictive performance of each optimized ML model for multi-target prediction, Fig. 5 described the plots of predicted and actual values. It could be found that the optimal GBR and RF can fit the training dataset effectively with the average train $R^2 > 0.92$ for the three targets and also exhibited satisfactory performance (average $R^2 > 0.84$) for the testing data (Table 3). Furthermore, for different datasets, the optimized ML models expressed better performance from datasets #1, #3, and #4 than #2. These results seem satisfactory as the prediction performance was much better than that obtained in similar previous research: poor testing performance (average R^2 0.75 and average RMSE 4.69) of RF was obtained for predicting HTL Yield_oil, N_oil, and energy

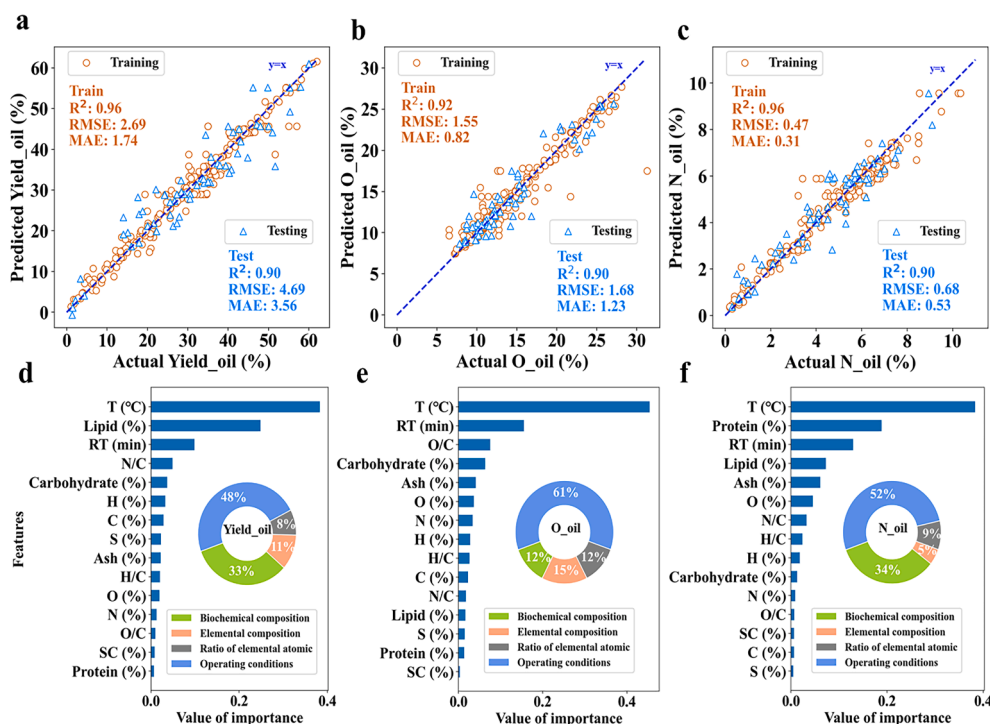


Fig. 4. Single-task prediction figures (training and testing) of Yield_oil (a), O_oil (b), and N_oil (c), and feature importance plots on Yield_oil (d), O_oil (e), and N_oil (f) based on optimal GBR from dataset #4.

Table 3

Summary of results from hyper-parameters adjusting and model performance of multi-task ML prediction.

Dataset	Input variables	Target	Optimizedmodel	Train					Test					Avg _b	MAE	Avg _b
				R^2	Avg _b	RMSE	Avg _b	MAE	R^2	Avg _b	RMSE	Avg _b	MAE			
#1	EC + Operating conditions	Yield_oil	GBR (128, 4) ^a	0.96	0.95	2.57	1.52	1.68	0.93	0.88	5.45	2.64	3.96	1.92		
		O_oil		0.91		1.57		0.84		0.89	1.70		1.22			
		N_oil	RF (32, 16)	0.97		0.41		0.27		0.88	0.77		0.56			
		Yield_oil		0.95	0.93	2.99	1.73	2.03	1.11	0.86	6.06	2.89	4.52	2.15		
		O_oil		0.90		1.67		0.95		0.87	2.89		1.35			
#2	Ratios + Operating conditions	N_oil	GBR (128, 4)	0.94		0.53		0.34		0.87	0.78		0.57			
		Yield_oil		0.96	0.95	2.66	1.55	1.79	0.97	0.83	6.60	3.06	4.33	2.06		
		O_oil	RF (32, 16)	0.91		1.58		0.87		0.88	1.78		1.39			
		N_oil		0.97		0.41		0.26		0.86	0.81		0.60			
		Yield_oil		0.95	0.93	2.98	1.72	2.06	1.12	0.80	7.08	3.30	4.95	2.35		
#3	BC + Operating conditions	O_oil	GBR (100, 4)	0.90		1.68		0.96		0.85	2.05		1.52			
		N_oil		0.95		0.50		0.32		0.87	0.78		0.58			
		Yield_oil	RF (32, 16)	0.96	0.94	2.67	1.58	1.83	1.02	0.88	5.53	2.69	3.93	1.95		
		O_oil		0.91		1.62		1.02		0.87	1.86		1.39			
		N_oil		0.96		0.44		0.28		0.89	0.69		0.53			
#4	EC + BC + Ratios + Operating conditions	Yield_oil	GBR (100, 4)	0.96	0.95	2.61	1.54	1.74	0.97	0.88	5.52	2.63	3.62	1.80		
		O_oil		0.91		1.60		0.90		0.89	1.70		1.24			
		N_oil	RF (32, 8)	0.97		0.41		0.26		0.90	0.68		0.54			
		Yield_oil		0.97	0.92	3.17	1.87	2.38	1.33	0.87	5.72	2.77	4.33	2.10		
		O_oil		0.88		1.83		1.18		0.87	0.85		1.41			
		N_oil		0.93		0.61		0.43		0.89	0.74		0.56			

EC: Elementary Composition (C, H, O, N and S); Ratios: Ratio of Elementary Composition (H/C, O/C, and N/C); BC: Biochemical Composition (Protein, Lipid; Carbohydrate and Ash); Yield_oil: The yield of bio-oil; O_oil: Oxygen Content in bio-oil; N_oil: Nitrogen Content in bio-oil; + denotes the combination of input variables under consideration.

^a The optimal hyper-parameters of ML models (n_estimators, max_depth).

^b Avg: Average values of training or testing performance for three targets.

recovery (Li et al., 2021c). The reason maybe because of the better generalization ability of models in this study as only algal biomass was used.

3.3.2. Model assessment and feature analysis with the partial dependence plots

The predicted values vs. actual experimental values for each output target in optimized GBR models with datasets #3 and #4 show a more

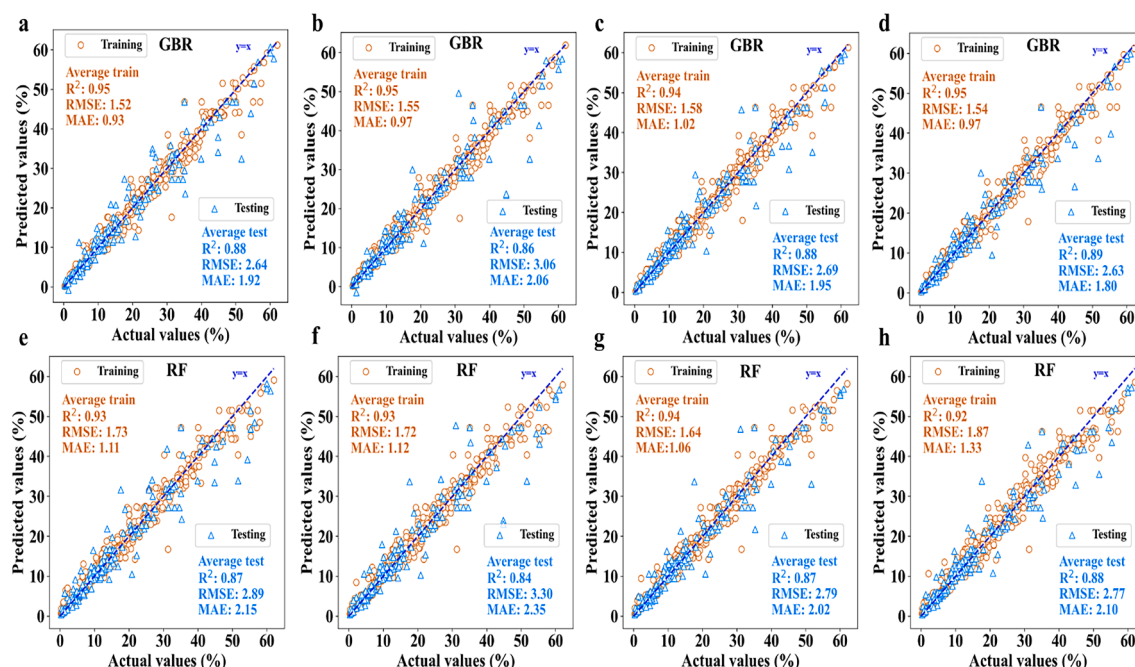


Fig. 5. Multi-task prediction figures (training and testing) from datasets #1 (a, e), #2 (b, f), #3 (c, g), and #4 (d, h) based on optimal GBR and RF.

detailed outlook of Fig. 5c and d (see [supplementary material](#)). The actual Yield_oil distributed in a wide range from 1% to 60%. The test R^2 value of Yield_oil in multi-task prediction (0.88, Table 3) was slightly lower than single-task (0.90, Table 2), while the test RMSE and MAE values were higher than those of the single-target prediction, respectively. For O_oil, the values ranged from 5% to 30% and concentrated between 6% and 25%. The test performance was lower than Yield_oil and N_oil prediction. The actual values of N_oil ranged 0–10% and centralized within 4% and 8%, and the testing performance was better than Yield_oil and N_oil prediction. In summary, both the two models with single-task or multi-task prediction for all targets exhibited satisfactory results with training $R^2 > 0.9$ and testing $R^2 \sim 0.90$ (Tables 2 and 3). Additionally, the accuracy of ML models for multi-task prediction and considering different input variables and the size of the dataset were competitive compared to related ML models published recently (with 229–448 datasets and test R^2 0.69–0.94) on this HTL topic (Gopirajan et al., 2021; Li et al., 2021c).

The T was the most critical feature to Yield_oil, O_oil, and N_oil together, followed by lipid and RT as second and third important features (see [supplementary material](#)). Other variables seem less important than the three features to overall three targets with the value of relative importance < 0.1 . The partial dependence plots of top-three vital input features show the impact of each variable on targets (see [supplementary material](#)). The Yield_oil increased to the peak at around 370 °C and then decreased slowly because supercritical gasification becomes dominant gradually at higher temperatures. Similarly, the bio-oil yield increased and then declined with increasing RT.

Moreover, the lipid content was positively related to the Yield_oil, and they expressed a near-linear relation in the lipid range of 10–25%, which was agreed with previous literature (Leow et al., 2015; Li et al., 2017). During HTL of microalgae at intermediate T of 350 °C, the maximum Yield_oil was achieved when RT of 15 min (Christensen et al., 2014). These results could show that high-lipid algae could produce a high bio-oil under a median T and relatively short RT condition.

The O_oil decreased with the increase of T (see [supplementary material](#)) due to enhancing the deoxygenation reactions at higher T (Zeb et al., 2017). RT had a similar effect on O_oil (see [supplementary material](#)). However, the O_oil remained stable between the O/C ratio of 0.10 and 0.75, but it increased sharply when the O/C ratio was higher

than 0.75 (see [supplementary material](#)). Therefore, it was vital to use algae of a low O/C ratio to obtain low O_oil, in addition to relatively high T and short RT.

For N_oil, T, protein, and RT are top-three critical inputs (Fig. 4). Algae, as a kind of high-protein biomass, N-containing compounds were enriched in the oil phase during HTL. The increasing T and extending RT mean higher reaction severity (Gollakota and Savage, 2020), and it promotes the conversion of algae to bio-oil, including N-containing bio-oil. Furthermore, the generation of heterocyclic nitrogenates in bio-oil stabilized during HTL at supercritical conditions (Leng et al., 2020c). Therefore, the N_oil may maintain a high value even at high temperatures (see [supplementary material](#)). On the other hand, it was found that algae with low protein reacted at low T, and RT could produce lower N_oil.

3.4. Optimization based on machine learning to guide experiments

3.4.1. Reverse optimization and experimental verification with model compounds

The reasonable optimization schemes and predicted values were obtained with PSO from the optimized GBR model with dataset #3 (Table 4). Then, corresponding experimental validations were conducted based on the optimized biochemical composition and HTL operating parameter values. The most experimental results of Yield_oil and N_oil agreed with the predicted ones except the O_oil (Table 4). Note that the O_oil was a value obtained by difference (Eq. 3), and other elements (C_oil, H_oil, and N_oil) could produce errors that may magnify the O_oil errors gradually. During HTL with low T, long RT, and high SC, the Yield_oil, O_oil, and N_oil can reach 66.46%, 14.38%, and 2.73%, respectively (Run 1, Table 4). Surprisingly, the modeling predictive result was well experimentally verified in Run 1 with $< 4\%$ errors for Yield_oil and N_oil, and 22% for O_oil. It noted mentioning that the optimal solution of lipid content (57.08%) from this optimization was close to the maximum value (58.70%, see [supplementary material](#)). However, the optimal bio-oil properties (maximum yield and minimum O and N) can also be obtained with feedstock of lower lipid (48.60%, Run 3 in Table 4). At T around 370 °C, the predicted Yield_oil and N_oil were comparable to those in HTL at lower temperatures (250–308 °C, Runs 1–3), while the O_oil was lower. In summary, the results from these

Table 4

The optimal solution and corresponding predicted and experimental outputs from ML models.

Optimal solution	Protein, Lipid, Carbohydrate, % ^a	T, °C	RT, min	SC, %	Yield_oil, %	O_oil, % ^c	N_oil, %
Reverse optimization with model compounds based on GBR developed with dataset #3.							
Run 1	17.37, 57.08, 22.85	247.00	87.00	27.00	64.00(66.46) ^b	11.75 (14.38)	2.82 (2.73)
Run 2	22.20, 58.70, 10.60	259.40	34.00	5.20	56.87 (65.90)	11.80 (13.27)	2.80 (2.21)
Run 3	27.80, 48.60, 22.20	308.00	68.00	27.00	59.71 (64.10)	9.14 (12.52)	3.34 (3.22)
Run 4	19.32, 55.75, 24.13	368.00	43.20	7.60	58.20 (56.61)	7.60 (17.72)	1.89 (2.01)
Run 5	20.12, 55.90, 14.20	373.00	84.00	5.00	58.13 ^d	7.84	2.24
Run 6	20.20, 58.70, 20.10	373.00	96.00	15.00	57.23	8.06	2.01
Run 7	38.90, 48.57, 11.53	373.00	102.00	29.80	60.30	8.09	3.42
Run 8	33.60, 47.80, 18.50	375.00	72.00	27.60	60.00	8.4	2.24
Optimization of bio-oil production from <i>Chlorella</i> based on GBR developed with dataset #4							
Run 9	65.32, 11.06, 14.35	240	2	10	23.87 (22.39)	23.94	3.70
Run 10		240	31	10	29.47 (29.60)	14.75 (14.10)	4.30 (7.99)
Run 11		260	31	5	30.17 (32.80)	14.78 (14.22)	5.19 (8.44)
Run 12		260	31	10	29.47 (30.96)	14.75	5.26
Run 13		300	2	5	32.29 (34.22)	19.31 (12.69)	5.00 (7.92)
Run 14		300	2	10	31.51 (31.14)	19.34 (14.78)	5.20 (8.52)
Run 15		370	2	5	38.35 (25.20)	17.91 (12.54)	4.92 (5.42)
Run 16		370	10	10	40.11 (23.45)	11.17 (14.80)	6.11 (5.78)

^a Ash = 100 – Protein – Lipid – Carbohydrate.^b Predicted and experimental values, and the latter is shown in the parentheses.^c By difference, O_oil = 100 – C_oil – H_oil – N_oil, daf basis.^d Not experimentally verified, and only predicted value was provided.

four runs achieved the desirable experimental verifications of the GBR model developed with dataset #3. Therefore, with the application of ML and reverse optimization, the HTL operating conditions and biochemical composition of feedstocks can be optimized simultaneously.

3.4.2. Forward optimization to tailor HTL conditions for specified algae biomass with bio-oil production

Since the feature importance of SC was not prominent for bio-oil compared to other operating conditions, the SC was offset at 5% or 10% in the forward optimization. The T and RT were optimized to achieve the optimal HTL operating conditions of bio-oil production with *Chlorella* based on GBR developed with dataset #4 (Fig. 6). From the outputs, eight optimal solutions (Yield_oil > 23.8%, N_oil < 6.2%, O_oil < 24.0%) with different operating conditions was chose to conduct experimental verification. Table 4 exhibits the results of optimal solutions (screened from Fig. 6) and experiment output. For most cases, Yield_oil prediction can be well verified with experiments except Runs 15 and 16. At T 240–300 °C (Runs 9–14), the error of Yield_oil between experiment and prediction was desirable (<8%). However, at 370 °C, the predicted bio-oil yield was > 38%, meaning possible overestimation of Yield_oil (experimental values < 26%, Table 5). This is quite interesting as a similar run (Run 4 in Table 4) was experimentally verified desirably for Yield_oil. The differences between reverse and forward

optimization experimental verifications may be caused by the variation of the feedstock used for HTL, but the difference between ML models for datasets #3 and #4 may also be a possible reason. Furthermore, the errors (<5%) between experimental and predictive results of O_oil were acceptable (Run 10 and 11), while the practical values of N_oil in most runs (Runs 10, 11, 13, and 14) were significantly higher than the predicted ones. However, at higher T 370 °C, the predictive values of N_oil and O_oil well agreed with the experiments (Runs 15 and 16). Again, the differences between the reverse and forward verifications maybe because of feedstock variation and model differences. To some extent, the bad performance on validation of N_oil in the forward verification might be caused by ML model overfitting (hyper-parameter tuning of ML models for prediction of N_oil had a low average R^2 of 0.67, Fig. 3) for different reasons, such as unrepresentative sample data and too many redundant input features in the limited dataset #4.

Based on the results from the reverse and forward optimization experimental verifications, it can be concluded that the experimental verifications may not necessarily be desirable (for example, errors > 35% for some Yield_oil and N_oil in this study) although the prediction performance seems favorable (for example, R^2 within 0.84–0.90). Maybe using more algal species (or algal mixtures) for HTL experimental verification can help compromise the difference caused by biomass variation and increase the verification performance. Additionally, a

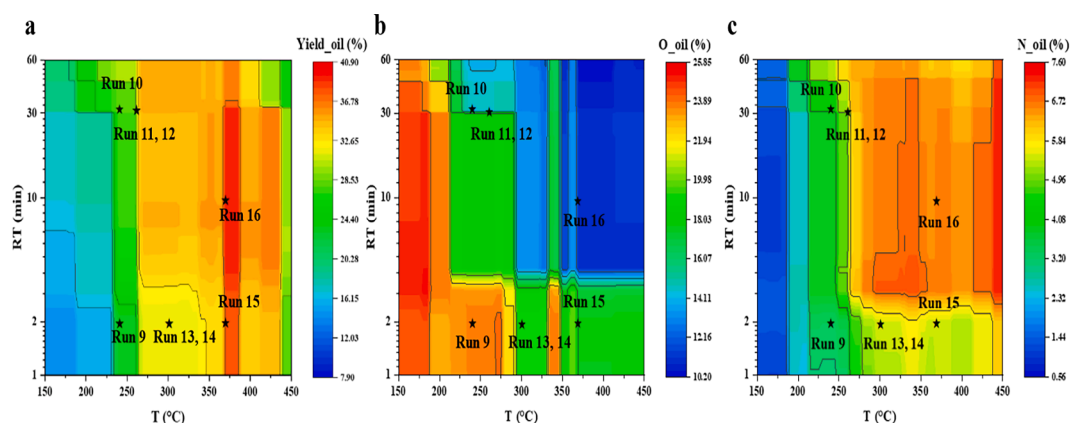


Fig. 6. The optimal operating conditions (the black stars in the plots) for bio-oil production with *Chlorella* of maximized Yield_oil (a), minimized O_oil (b), and N_oil (c) based on dataset #4.

sizeable dataset with heating rate, pressure, catalysts, solvents, and separation order of bio-oil could be compiled to improve the generalization and optimization ability of models in future ML-based HTL studies.

3.5. Implication and potential insights

In this work, we provide a comprehensive modeling-coupled experiment investigation framework based on the HTL of algae for desired bio-oil production. Concerning the HTL process, several scientific merits can be achieved. Firstly, the developed ML model can predict the bio-oil properties, including the yield, N, and O content of oil with specific feedstock and operational conditions as inputs. The predicted properties can be used to evaluate whether a newly provided feedstock is suitable for HTL to produce bio-oil. Moreover, the predicted bio-oil properties can provide meaningful information for determining bio-oil application or downstream biocrude upgrading strategies. Secondly, the insights from model-based feature importance and correlation analysis can help us identify critical factors and understand their impacts on bio-oil yield and properties, which benefit us to design a new experimental setup to control these important factors in the right direction to produce bio-oil with desired properties. Thirdly, the ML-based optimization modeling method can provide ideal feedstock composition ranges and new experimental designs for desired targets. Researchers can select suitable feedstock or mix multiple feedstocks based on the ideal ranges of biochemical compositions, or our developed model can provide experimental conditions for a specific feedstock. This kind of model-guided experiment helps simplify the experimental investigation procedure and requires less labor, time, energy, and cost.

Apart from the HTL case in the present work, the comprehensive modeling-coupled experiment investigation framework is a type of one-round active learning process which is to reach the purpose by conducting experiment-model-experiment iterations to generate more and more data for enhancing the machine learning model and finally achieving a desired product or material. Therefore, our model can be further improved by integrating more related new data points to improve the robustness and generalization and provide more accurate guidance for experimentalists. To sum up, as presented in this study, researchers in both academia and industry can employ the model-coupled experiment framework to aid process design, interpretation, and optimization for producing products with desired quality.

4. Conclusions

The yield, O, and N content of bio-oil were successfully predicted with the algae properties and operational conditions in HTL by ML predictive models. The GBR model was better for single-task and multi-task prediction with average training $R^2 > 0.9$ and testing $R^2 \sim 0.90$ compared to RF. The feature importance and partial dependence were analyzed, and T, lipid, and RT were the three crucial variables for the overall three output targets. The ML-based predictive optimal solutions were verified with reverse and forward optimal experiments, which obtained acceptable results but need further improvement.

CRedit authorship contribution statement

Weijin Zhang: Investigation, Writing – original draft, Formal analysis. **Jie Li:** Methodology, Writing – review & editing. **Tonggui Liu:** Data curation, Visualization. **Songqi Leng:** Data curation, Validation. **Lihong Yang:** Software. **Haoyi Peng:** Resources. **Shaojian Jiang:** Resources. **Wenguang Zhou:** Resources, Funding acquisition. **Lijian Leng:** Supervision, Conceptualization, Funding acquisition, Writing – review & editing. **Hailong Li:** Resources, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The work was supported by the National Key Research and Development Program of China (2021YFE0104900) and the National Natural Science Foundation of China (No. 51808278).

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.biortech.2021.126011>.

References

- Cheng, F., Porter, M.D., Colosi, L.M., 2020. Is hydrothermal treatment coupled with carbon capture and storage an energy-producing negative emissions technology? *Energy Convers. Manag.* 203, 112252. <https://doi.org/10.1016/j.enconman.2019.112252>.
- Christensen, P.S., Peng, G., Vogel, F., Iversen, B.B., 2014. Hydrothermal Liquefaction of the Microalgae *Phaeodactylum tricornutum*: Impact of Reaction Conditions on Product and Elemental Distribution. *Energy & Fuels* 28, 5792–5803. <https://doi.org/10.1021/ef5012808>.
- Dietterich, T.G., 2000. An Experimental Comparison of Three Methods for Constructing Ensembles of Decision Trees: Bagging, Boosting, and Randomization. *Mach. Learn.* 139–158.
- Fan, L., Zhang, H., Li, J., Wang, Y., Leng, L., Li, J., Yao, Y., Lu, Q., Yuan, W., Zhou, W., 2020. Algal biorefinery to value-added products by using combined processes based on thermochemical conversion: A review. *Algal Res.* 47, 101819. <https://doi.org/10.1016/j.algal.2020.101819>.
- Gollakota, Akhila, Savage, Phillip E., 2020. Fast and Isothermal Hydrothermal Liquefaction of Polysaccharide Feedstocks. *ACS Sustain. Chem. Eng.* 8 (9), 3762–3772. <https://doi.org/10.1021/acssuschemeng.9b06873>.
- Gopirajan, P.V., Gopinath, K.P., Sivarajani, G., Arun, J., 2021. Optimization of hydrothermal liquefaction process through machine learning approach: process conditions and oil yield. *Biomass Convers. Biorefinery*. <https://doi.org/10.1007/s13399-020-01233-8>.
- Huang, H., Yuan, X., 2015. Recent progress in the direct liquefaction of typical biomass. *Prog. Energy Combust. Sci.* 49, 59–80. <https://doi.org/10.1016/j.pecs.2015.01.003>.
- Juneja, M., Nagar, S.K., 2016. Particle swarm optimization algorithm and its parameters: A review. In: in: 2016 International Conference on Control, Computing, Communication and Materials (ICCCCM). IEEE, pp. 1–5. <https://doi.org/10.1109/ICCCCM.2016.7918233>.
- Leng, L., Li, H., Yuan, X., Zhou, W., Huang, H., 2018a. Bio-oil upgrading by emulsification/microemulsification: A review. *Energy* 161, 214–232. <https://doi.org/10.1016/j.energy.2018.07.117>.
- Leng, L., Li, J., Wen, Z., Zhou, W., 2018b. Use of microalgae to recycle nutrients in aqueous phase derived from hydrothermal liquefaction process. *Bioresour. Technol.* 256, 529–542. <https://doi.org/10.1016/j.biortech.2018.01.121>.
- Leng, L., Li, Jun, Yuan, X., Li, Jingjing, Han, P., Hong, Y., Wei, F., Zhou, W., 2018c. Beneficial synergistic effect on bio-oil production from co-liquefaction of sewage sludge and lignocellulosic biomass. *Bioresour. Technol.* 251, 49–56. <https://doi.org/10.1016/j.biortech.2017.12.018>.
- Leng, Lijian, Wei, Liang, Xiong, Qin, Xu, Siyu, Li, Wenting, Lv, Sen, Lu, Qian, Wan, Liping, Wen, Zhiyou, Zhou, Wenguang, 2020a. Use of microalgae based technology for the removal of antibiotics from wastewater: A review. *Chemosphere* 238, 124680. <https://doi.org/10.1016/j.chemosphere.2019.124680>.
- Leng, Lijian, Xu, Siyu, Liu, Renfeng, Yu, Ting, Zhuo, Ximeng, Leng, Songqi, Xiong, Qin, Huang, Huajun, 2020b. Nitrogen containing functional groups of biochar: An overview. *Bioresour. Technol.* 298, 122286. <https://doi.org/10.1016/j.biortech.2019.122286>.
- Leng, L., Yuan, X., Chen, X., Huang, H., Wang, H., Li, H., Zhu, R., Li, S., Zeng, G., 2015. Characterization of liquefaction bio-oil from sewage sludge and its solubilization in diesel microemulsion. *Energy* 82, 218–228. <https://doi.org/10.1016/j.energy.2015.01.032>.
- Leng, Lijian, Zhang, Weijin, Peng, Haoyi, Li, Hailong, Jiang, Shaojian, Huang, Huajun, 2020c. Nitrogen in bio-oil produced from hydrothermal liquefaction of biomass: A review. *Chem. Eng. J.* 401, 126030. <https://doi.org/10.1016/j.cej.2020.126030>.
- Leng, Songqi, Li, Wenting, Han, Chao, Chen, Linlin, Chen, Jie, Fan, Liangliang, Lu, Qian, Li, Jun, Leng, Lijian, Zhou, Wenguang, 2020d. Aqueous phase recirculation during hydrothermal carbonization of microalgae and soybean straw: A comparison study. *Bioresour. Technol.* 298, 122502. <https://doi.org/10.1016/j.biortech.2019.122502>.
- Leow, Shijie, Witter, John R., Vardon, Derek R., Sharma, Brajendra K., Guest, Jeremy S., Strathmann, Timothy J., 2015. Prediction of microalgae hydrothermal liquefaction

- products from feedstock biochemical composition. *Green Chem.* 17 (6), 3584–3599. <https://doi.org/10.1039/C5GC00574D>.
- Li, J., Pan, L., Suvana, M., Tong, Y.W., Wang, X., 2020a. Machine Learning Prediction of Syngas Composition of Hydrothermal Gasification from Wet Organic Wastes. *Energy-proceedings*.
- Li, Jie, Pan, Lanjia, Suvana, Manu, Tong, Yen Wah, Wang, Xiaonan, 2020b. Fuel properties of hydrochar and pyrochar: Prediction and exploration with machine learning. *Appl. Energy* 269, 115166. <https://doi.org/10.1016/j.apenergy.2020.115166>.
- Li, Jie, Pan, Lanjia, Suvana, Manu, Wang, Xiaonan, 2021a. Machine learning aided supercritical water gasification for H₂-rich syngas production with process optimization and catalyst screening. *Chem. Eng. J.* 426, 131285. <https://doi.org/10.1016/j.cej.2021.131285>.
- Li, Jie, Suvana, Manu, Pan, Lanjia, Zhao, Yingru, Wang, Xiaonan, 2021b. A hybrid data-driven and mechanistic modelling approach for hydrothermal gasification. *Appl. Energy* 304, 117674. <https://doi.org/10.1016/j.apenergy.2021.117674>.
- Li, Jie, Wang, Zhiyong, Yang, Xi, Hu, Ling, Liu, Yuwen, Wang, Cunxin, 2007. Evaluate the pyrolysis pathway of glycine and glycyglycine by TG-FTIR. *J. Anal. Appl. Pyrolysis* 80 (1), 247–253. <https://doi.org/10.1016/j.jaap.2007.03.001>.
- Li, Jie, Zhang, Weijin, Liu, Tonggui, Yang, Lihong, Li, Hailong, Peng, Haoyi, Jiang, Shaojian, Wang, Xiaonan, Leng, Lijian, 2021c. Machine learning aided bio-oil production with high energy recovery and low nitrogen content from hydrothermal liquefaction of biomass with experiment verification. *Chem. Eng. J.* 425, 130649. <https://doi.org/10.1016/j.cej.2021.130649>.
- Li, Jie, Zhu, Xinzhe, Li, Yinan, Tong, Yen Wah, Ok, Yong Sik, Wang, Xiaonan, 2021d. Multi-task prediction and optimization of hydrochar properties from high-moisture municipal solid waste: Application of machine learning on waste-to-resource. *J. Clean. Prod.* 278, 123928. <https://doi.org/10.1016/j.jclepro.2020.123928>.
- Li, Yalin, Leow, Shijie, Fedders, Anna C., Sharma, Brajendra K., Guest, Jeremy S., Strathmann, Timothy J., 2017. Quantitative multiphase model for hydrothermal liquefaction of algal biomass. *Green Chem.* 19 (4), 1163–1174. <https://doi.org/10.1039/C6GC03294J>.
- Lin, Yuan, Wu, Jiajin, Xu, Bo, Xu, Kan, Lin, Hongfei, 2019. Learning to rank using multiple loss functions. *Int. J. Mach. Learn. Cybern.* 10 (3), 485–494. <https://doi.org/10.1007/s13042-017-0730-4>.
- Liu, An-An, Su, Yu-Ting, Nie, Wei-Zhi, Kankanhalli, Mohan, 2017. Hierarchical Clustering Multi-Task Learning for Joint Human Action Grouping and Recognition. *IEEE Trans. Pattern Anal. Mach. Intell.* 39 (1), 102–114. <https://doi.org/10.1109/TPAMI.3410.1109/TPAMI.2016.2537337>.
- Madsen, R.B., Bernberg, R.Z.K., Biller, P., Becker, J., Iversen, B.B., Glasius, M., 2017. Hydrothermal co-liquefaction of biomasses – quantitative analysis of bio-crude and aqueous phase composition. *Sustain. Energy Fuels* 1, 789–805. <https://doi.org/10.1039/C7SE00104E>.
- Natekin, A., Knoll, A., 2013. Gradient boosting machines, a tutorial. *Front. Neurobot.* 7 <https://doi.org/10.3389/fnbot.2013.00021>.
- Pathy, A., Meher, S., P, B., 2020. Predicting algal biochar yield using eXtreme Gradient Boosting (XGB) algorithm of machine learning methods. *Algal Res.* 50, 102006. <https://doi.org/10.1016/j.algal.2020.102006>.
- Peterson, A.A., Vogel, F., Lachance, R.P., Fröling, M., Antal Jr., M.J., Tester, J.W., 2008. Thermochemical biofuel production in hydrothermal media: A review of sub- and supercritical water technologies. *Energy Environ. Sci.* 1, 32. <https://doi.org/10.1039/b810100k>.
- Qi, C., Chen, Q., 2018. Evolutionary Random Forest Algorithms for Predicting the Maximum Failure Depth of Open Stope Hangingwalls. *IEEE Access* 6, 72808–72813. <https://doi.org/10.1109/ACCESS.2018.2880009>.
- Qiao, Y., Wang, B., Ji, Y., Xu, F., Zong, P., Zhang, J., Tian, Y., 2019. Thermal decomposition of castor oil, corn starch, soy protein, lignin, xylan, and cellulose during fast pyrolysis. *Bioresour. Technol.* 278, 287–295. <https://doi.org/10.1016/j.biortech.2019.01.102>.
- Savage, P.E., 2012. Algae Under Pressure and in Hot Water. *Science* (80–). 338, 1039–1040. <https://doi.org/10.1126/science.1224310>.
- Sheng, L., Wang, X., Yang, X., 2018. Prediction model of biocrude yield and nitrogen heterocyclic compounds analysis by hydrothermal liquefaction of microalgae with model compounds. *Bioresour. Technol.* 247, 14–20. <https://doi.org/10.1016/j.biortech.2017.08.011>.
- Sohail, S., Rosendahl, L., Rudolf, A., 2011. Hydrothermal liquefaction of biomass : A review of subcritical water technologies. *Energy* 36, 2328–2342. <https://doi.org/10.1016/j.energy.2011.03.013>.
- Song, Wenhan, Wang, Shuzhong, Xu, Donghai, Guo, Yang, Yang, Chuang, Zhang, Jiandong, Li, Yanhui, 2019. Comprehensive potential evaluation of the bio-oil production and nutrient recycling from seven algae through hydrothermal liquefaction. *Korean J. Chem. Eng.* 36 (10), 1604–1618. <https://doi.org/10.1007/s11814-019-0345-4>.
- Tang, Qinghui, Chen, Yingquan, Yang, Haiping, Liu, Ming, Xiao, Haoyu, Wu, Ziyue, Chen, Hanping, Naqvi, Salman Raza, 2020. Prediction of Bio-oil Yield and Hydrogen Contents Based on Machine Learning Method: Effect of Biomass Compositions and Pyrolysis Conditions. *Energy & Fuels* 34 (9), 11050–11060. <https://doi.org/10.1021/acs.energyfuels.0c01893>.
- Xu, Siyu, Chen, Jiefeng, Peng, Haoyi, Leng, Songqi, Li, Hui, Qu, Wenqi, Hu, Yingchao, Li, Hailong, Jiang, Shaojian, Zhou, Wenguang, Leng, Lijian, 2021. Effect of biomass type and pyrolysis temperature on nitrogen in biochar, and the comparison with hydrochar. *Fuel* 291, 120128. <https://doi.org/10.1016/j.fuel.2021.120128>.
- Yang, Jie, He, Quan (Sophia), Niu, Haibo, Corscadden, Kenneth, Astatkie, Tess, 2018. Hydrothermal liquefaction of biomass model components for product yield prediction and reaction pathways exploration. *Appl. Energy* 228, 1618–1628. <https://doi.org/10.1016/j.apenergy.2018.06.142>.
- Zeb, H., Park, J., Riaz, A., Ryu, C., Kim, J., 2017. High-yield bio-oil production from macroalgae (*Saccharina japonica*) in supercritical ethanol and its combustion behavior. *Chem. Eng. J.* 327, 79–90. <https://doi.org/10.1016/j.cej.2017.06.078>.
- Zhu, Xinzhe, Li, Yinan, Wang, Xiaonan, 2019. Machine learning prediction of biochar yield and carbon contents in biochar based on biomass characteristics and pyrolysis conditions. *Bioresour. Technol.* 288, 121527. <https://doi.org/10.1016/j.biortech.2019.121527>.